Runming Yang

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Ph.D. student at the University of Hong Kong working on efficient LLM post-training, knowledge distillation, and low-bit quantization.

Education

The University of Hong Kong Oct. 2025 - Present

Ph.D. in Electrical and Electronic Engineering

Expected graduation: Sep. 2029; Advisor: Prof. Ngai Wong; Research: LLM efficiency and post-training

Tsinghua University Sep. 2022 - Jun. 2025

M.Eng. in Big Data Engineering

Advisor: Prof. Yujiu Yang; GPA: 3.73/4.00; Research: model compression and knowledge distillation

Huazhong University of Science and Technology Sep. 2017 - Jun. 2021

B.Eng. in AI and Automation

Science and Engineering Cross-Innovation Experimental Class

Research Highlights

Shadow-FT: transferring fine-tuning updates Equal contribution

Studies weight similarity between paired base and instruction-tuned models. Fine-tuning updates learned on the base model are transferred to its instruction-tuned counterpart.

PTQTP: low-bit LLM inference Equal contribution

Decomposes weights into two ternary trit-planes for multiplication-free inference. Reports 4.63-fold higher end-to-end generation throughput than FP16 for LLaMA-2 7B on an RTX 3090 (batch size 1; 512 input and 512 generated tokens).

LoCa: logit calibration for distillation First author

Calibrates teacher logits for misclassified samples to improve knowledge distillation with limited additional computation.

Industry Experience

Tencent PCG Nov. 2023 - Jun. 2025

Algorithm Intern, Social Secure Center

- Built domain-specific multimodal datasets for model development and evaluation.
- Conducted model post-training and evaluation on domain-specific tasks.
- Investigated cross-modal feature fusion for multimodal modeling.

Publications and Manuscripts

* indicates equal contribution.

Publications

1. Revisiting Model Interpolation for Efficient Reasoning
T. Wu, **R. Yang**, T. Liu, J. Wang, N. Wong
ACL 2026 (Main) Paper · Code

2. ProFit: Leveraging High-Value Signals in SFT via Probability-Guided Token Selection
T. Liu*, T. Wu*, **R. Yang**, S. Sun, J. Wang, Y. Yang
ACL 2026 (Findings) Paper · Code

3. Can We Trust LLMs on Memristors? Diving into Reasoning Ability under Non-Ideality
T. Wu*, Y. Cheng*, C. Ding*, **R. Yang**, X. Feng, W. Zhou, Z. Liu, N. Wong
COLM 2026 Paper

4. Rethinking Kullback-Leibler Divergence in Knowledge Distillation for Large Language Models
T. Wu, C. Tao, J. Wang, **R. Yang**, Z. Zhao, N. Wong
COLING 2025 (Oral) Paper · Code

5. LoCa: Logit Calibration for Knowledge Distillation
R. Yang, T. Wu, Y. Yang
ECAI 2024 Paper

Preprints and Manuscripts

1. FluxBin: Flexible LUT-based Ultra-low-bit LLM Inference by Algorithm-Kernel Synergy
Q. Yang, **R. Yang**, H. Xiao, W. Xu, J. Chen, H. Liu, C. Ding, R. Hu, Y.-C. Wu, N. Wong
arXiv:2608.15602 (2026) Paper · Code

2. Shadow-FT: Tuning Instruct Model via Training on Paired Base Model
T. Wu*, **R. Yang***, J. Li, P. Hu, Y.-C. Wu, N. Wong, Y. Yang
arXiv:2505.12716 (2025) Paper · Code

3. Timber: Training-free Instruct Model Refining with Base via Effective Rank
T. Wu, **R. Yang**, T. Liu, J. Wang, Z. Xu, N. Wong
arXiv:2509.23595 (2025) Paper · Code

4. PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models
H. Xiao*, **R. Yang***, Q. Yang*, W. Xu, Z. Li, Y. Su, Z. Liu, H. Yang, N. Wong
arXiv:2509.16989 (2025) Paper · Project

5. Quantization Meets Reasoning: Exploring and Mitigating Degradation of Low-Bit LLMs in Mathematical Reasoning
Z. Li*, Y. Su*, S. Wang, **R. Yang**, C. Xie, A. Liu, M. Li, J. Cao, Y. Xie, N. Wong, H. Yang
arXiv:2505.11574 (2025) Paper